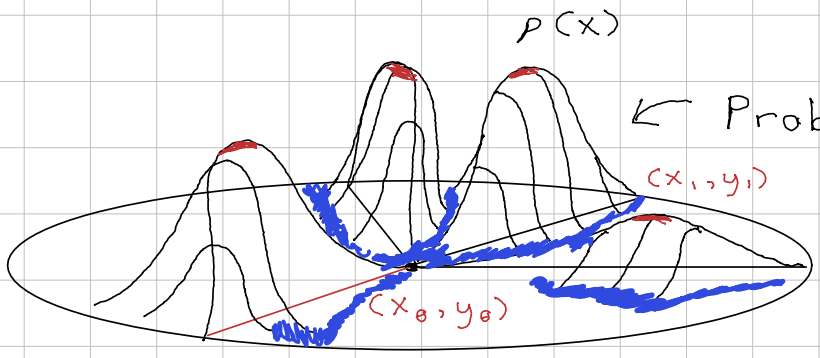
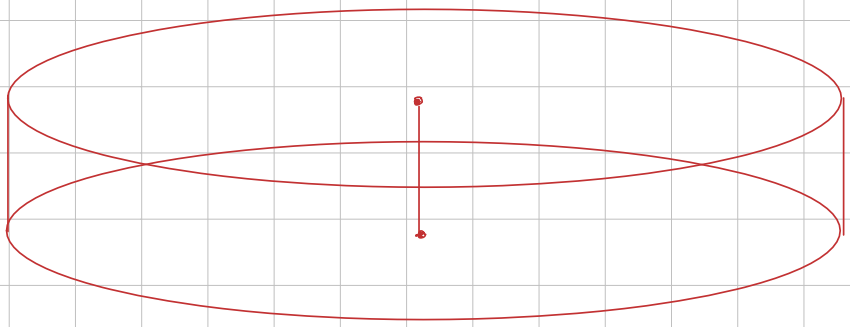


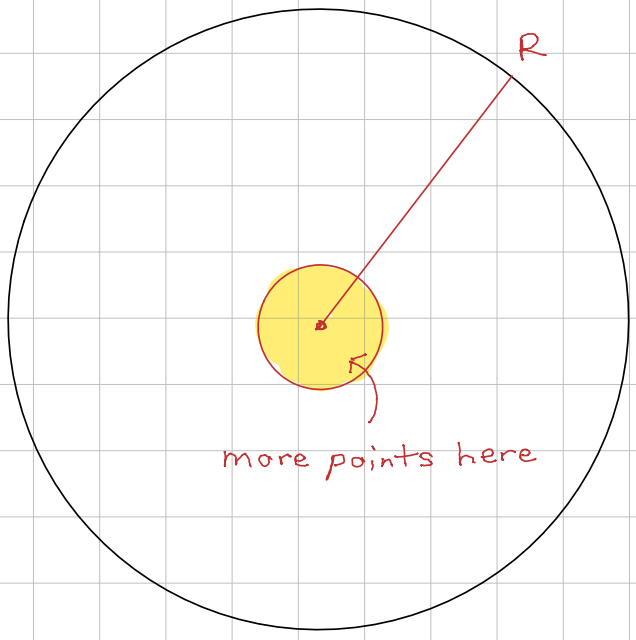
← Probability distribution over a territory

Non-hypothesis: $p(x)$ is a uniform distribution

- must disprove this to proceed, our research question

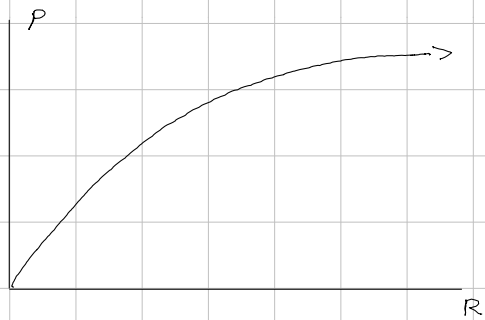


* events are located uniformly



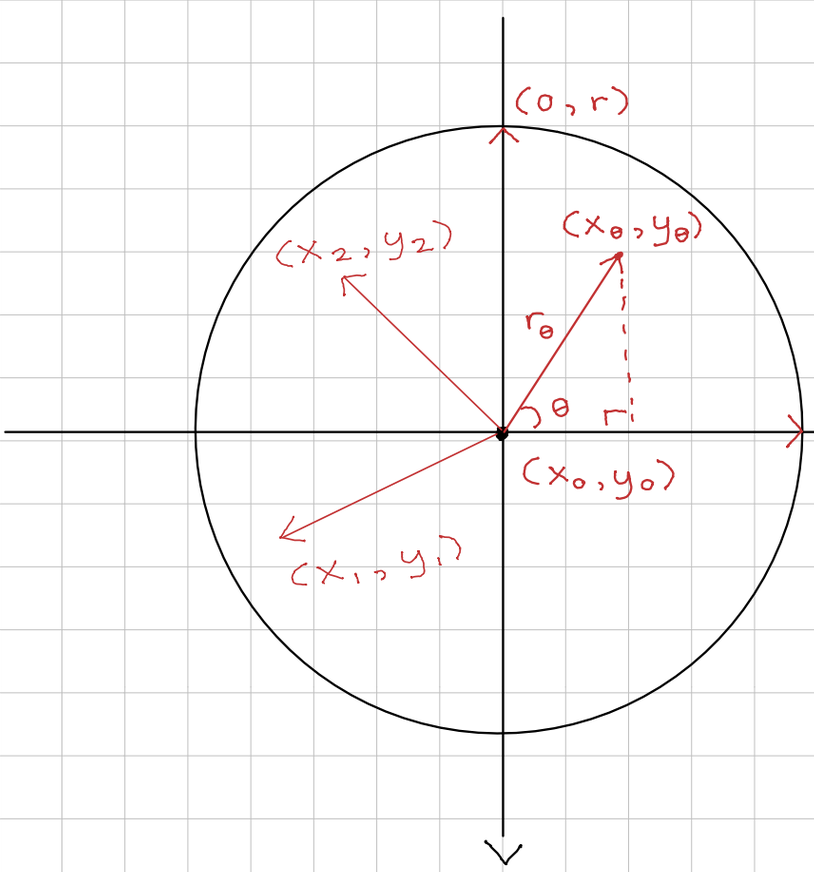
* as $r \to 0$, more clumping
* as $r \to R$, more empty space

So, apply a transformation to cluster outwards



algorithm: Pick a coordinate (x, y) at random to be center of territory

- From the (x, y), randomly choose a radius from a distribution to fix
- From that radius fixed, sample along that from a distribution
- Sample a θ from 0 to 2π according to the unit circle



* $r_\theta = 1$ for a unit circle

$\sin\theta = \frac{y_0}{r_\theta}$ $\quad \cos\theta = \frac{x_0}{r_\theta}$

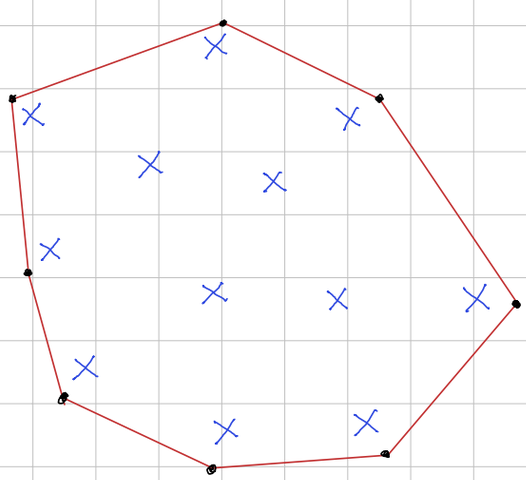
$\therefore y_0 = r_\theta \sin\theta + y_\theta$

$x_0 = r_\theta \cos\theta + x_\theta$

* radius, location, θ are all variable

* X_k & Y_k are dependent on them

- Once distribution of points plotted, run convex hull over points



* Need distributions of:
  - ovenbird territory sizes
  - average event densities
  - average number of territories in area